

Relaxation and Odd-Even Effect in Fermi Liquids

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Outline

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- k -space
- Fermi Dirac distribution
- Fermi surface

2

Recovering equations

- Linearization of the collision integral
- Linearized collision operator

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Odd-Even effect

1

Introduction

1.1

***k*-space**

k -space

- Analogous to momentum space
- Each point $\mathbf{k}$ in k -space corresponds to a wave $e^{i\mathbf{k}}$
- **Fourier transform** converts between k -space to real space

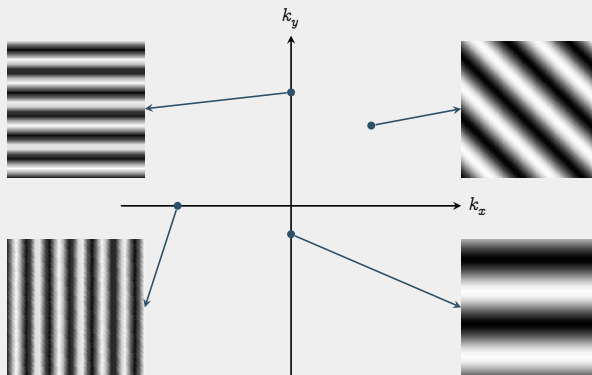


Figure 1: REPRESENTATIVE WAVES OF k -SPACE

- Distance from origin $\propto$ Wave frequency
- Direction on k -space $\implies$ Wave direction

1.2

Fermi Dirac distribution

$$f_0(\mathbf{p}) = \frac{1}{e^{\beta(\varepsilon(\mathbf{p}) - \mu)} + 1}$$

1

- $f_0(\mathbf{p})$ is the Fermi-Dirac distribution
- $\beta = 1/k_B T$ is the inverse temperature
- μ is the chemical potential

The Fermi-Dirac distribution is in equilibrium; the collision integral is zero.

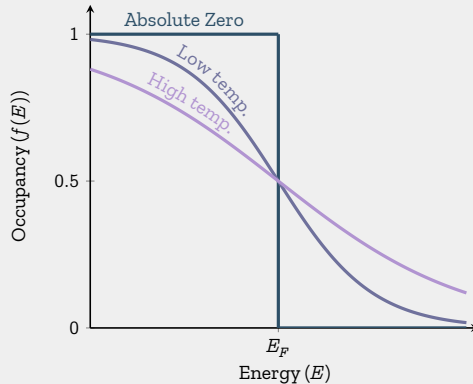


Figure 2: FERMI-DIRAC DISTRIBUTION AT DIFFERENT TEMPERATURES

- At $T = 0$, all states below the Fermi energy are filled, all states above are empty.
- As T increases, the distribution smooths out.

1.3

Fermi surface

The Fermi surface is the set of points in k -space where the energy of the electrons is equal to the Fermi energy.

Under no thermal gradients and external force, the Fermi surface is a a circle in 2D k -space.

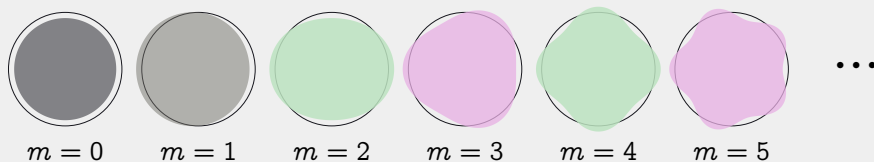


Figure 3: CIRCULAR HARMONICAL PERTURBATIONS OF THE FERMI SURFACE

The Fermi surface may be perturbed by external fields. Since a perturbation has to be smooth, we can expand the perturbation in terms of circular harmonics $e^{im\theta}$ in 2D.

2

Recovering equations

2.1

Linearization of the collision integral

Ideas

- Consider **only perturbation near Fermi's equilibrium**

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$$

- Transform the integrals into a linear operator

Preparation

Fermi equilibrium

$$f_0(\mathbf{p}) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1}$$

2

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{1}{T} f_0(1 - f_0)$$

3

$$\frac{\partial f_0}{\partial \mu} = \frac{1}{T} f_0(1 - f_0)$$

4

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{\partial f_0}{\partial \mu}$$

5

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$$

Perturbation

6

$$\approx f_0(\mathbf{p}) + \frac{\partial f_0}{\partial \mu} \delta \mu$$

Taylor series. Fix ε

7

$$\therefore \delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} \delta \mu$$

8

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$$

Perturbation

6

$$\approx f_0(\mathbf{p}) + \frac{\partial f_0}{\partial \mu} \delta \mu$$

Taylor series. Fix ε

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$$\therefore \delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} \delta \mu$$

8

Perturbation of interest

$$\delta \mu = T \psi$$

9

$\delta \mu$ represents temperature independent change in chemical potential

2.2

Linearized collision operator

$$\delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} T \psi \quad 10$$

$$= \left(-\frac{1}{T} f_0 (1 - f_0) \right) \times T \psi \quad 11$$

$$= f_0 (1 - f_0) \psi \quad 12$$

Collision integral

$$\hat{\mathcal{J}}[f] = - \int \dots \int \frac{1}{(2\pi)^6} d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) \\ [f(\mathbf{p}_1)f(\mathbf{p}_2)(1 - f(\mathbf{p}'_1))(1 - f(\mathbf{p}'_2)) - f(\mathbf{p}'_1)f(\mathbf{p}'_2)(1 - f(\mathbf{p}_1))(1 - f(\mathbf{p}_2))] \quad 13$$

3

Odd-Even effect